

■ 시나리오 별 백업 복구(User Managed) - Archive log mode

KEY WORDS

- <시나리오 14> 모든 데이터파일 손상되었을 경우 복구복구작업시 필요한 archive log file 삭제되었지만 redo log file 이 있기때문에 문제없이 복구
- <시나리오 15> 복구작업시 필요한 archive log file 이 손상되었을 경우 완전복구 실패로 인해 불완전 복구
- <시나리오 16> 데이터파일 삭제, 모든 아카이브파일이 삭제된 경우 불완전한 복구(일관성 있는 백업, open, online, hot backup 있음)
- <시나리오 17> 데이터파일 삭제 모든 아카이브 파일이 삭제된 경우 불완전 복구(일관성있는 백업, close, offline, cold)
- <시나리오 18> 데이터베이스가 정상적인 종료후에 inactive 상태에 있는 redo log file 삭제된 경우 복구
- <시나리오 19> 데이터베이스가 운영중에 inactive 상태에 있는 redo log file 삭제된 경우 복구

<<시나리오 14>> 모든 데이터파일 손상되었을 경우 복구복구작업시 필요한 archive log file 삭제되었지만 redo log file이 있기때문에 문제없이 복구

-- 데이터파일 온라인 백업(백업당시의 scn 번호와 sequence 번호를 저장)

```
SYS@orcl> alter database begin backup;
```

Database altered.

```
SYS@orcl> select a.file#, b.name tbs_name, a.name file_name, a.checkpoint_change#, a.status from v$datafile a, v$tablespace b where a.ts# = b.ts#;
```

FILE#	TBS_NAME	FILE_NAME	CHECKPOINT_CHANGE#	STATUS
1	SYSTEM	/u01/app/oracle/oradata/orcl/system01.dbf	2699840	SYSTEM
2	UNDOTBS1	/u01/app/oracle/oradata/orcl/undotbs01	2699840	ONLINE
3	SYSAUX	/u01/app/oracle/oradata/orcl/sysaux01.dbf	2699840	ONLINE
7	USERS	/u01/app/oracle/oradata/orcl/users01.dbf	2699840	ONLINE

```
SYS@orcl> ! cp -v /u01/app/oracle/oradata/orcl/*.dbf /home/oracle/backup/open_bkp/
'/u01/app/oracle/oradata/orcl/sysaux01.dbf' -> '/home/oracle/backup/open_bkp/sysaux01.dbf'
'/u01/app/oracle/oradata/orcl/system01.dbf' -> '/home/oracle/backup/open_bkp/system01.dbf'
'/u01/app/oracle/oradata/orcl/temp01.dbf' -> '/home/oracle/backup/open_bkp/temp01.dbf'
'/u01/app/oracle/oradata/orcl/users01.dbf' -> '/home/oracle/backup/open_bkp/users01.dbf'
```

```
SYS@orcl> alter database end backup;
Database altered.
```

```
SYS@orcl> alter database backup controlfile to '/home/oracle/backup/open_bkp/control_20250513.ctl';
Database altered.
```

```
Database altered.
```

```
SYS@orcl> select a.group#, b.sequence#, a.member, b.bytes/1024/1024 mb, b.status, b.archived, b.first_change#,
b.next_change# from v$logfile a, v$log b where a.group# = b.group# order by 1,2;
```

GROUP#	SEQUENCE#	MEMBER	MB
STATUS	ARC	FIRST_CHANGE#	NEXT_CHANGE#
-----	-----	-----	-----
	1	67 /u01/app/oracle/oradata/orcl/redo01.log	20 INACTIVE
YES	2595387	2595390	
	2	68 /u01/app/oracle/oradata/orcl/redo02.log	20 INACTIVE
YES	2595390	2698783	
	3	69 /u01/app/oracle/oradata/orcl/redo03.log	20 CURRENT
NO	2698783	1.8447E+19	

```
SYS@orcl> select a.file#, b.name tbs_name, a.name file_name, a.checkpoint_change#, a.status from v$datafile a,
v$tablespace b where a.ts# = b.ts#;
```

FILE#	TBS_NAME	FILE_NAME	CHECKPOINT_CHANGE#
STATUS			
-----	-----	-----	-----
	1 SYSTEM	/u01/app/oracle/oradata/orcl/system01.dbf	2699840
SYSTEM			
	2 UNDOTBS1	/u01/app/oracle/oradata/orcl/undotbs01	2699840 ONLINE
	3 SYSAUX	/u01/app/oracle/oradata/orcl/sysaux01.dbf	2699840
ONLINE			
	7 USERS	/u01/app/oracle/oradata/orcl/users01.dbf	2699840
ONLINE			

```
-- 백업이전 아카이브 삭제
```

```
SYS@orcl> !ls /home/oracle/arch*
```

/home/oracle/arch1:

```
0001_0000000062_1199981728.arc 0001_0000000065_1199981728.arc 0001_0000000068_1199981728.arc
0001_0000000063_1199981728.arc 0001_0000000066_1199981728.arc
0001_0000000064_1199981728.arc 0001_0000000067_1199981728.arc
```

/home/oracle/arch2:

```
0001_0000000062_1199981728.arc 0001_0000000065_1199981728.arc 0001_0000000067_1199981728.arc
0001_0000000063_1199981728.arc 0001_0000000066_1199981728.arc 0001_0000000068_1199981728.arc
```

```
SYS@orcl> !rm /home/oracle/arch{1,2}/*
```

```
SYS@orcl> alter system archive log current;
```

System altered.

```
SYS@orcl> alter system switch logfile;
```

System altered.

```
SYS@orcl> /
```

```
SYS@orcl> /
```

```
SYS@orcl> /
```

```
SYS@orcl> /
```

```
SYS@orcl> !ls /home/oracle/arch*
```

/home/oracle/arch1:

```
0001_0000000069_1199981728.arc 0001_0000000071_1199981728.arc 0001_0000000073_1199981728.arc
0001_0000000070_1199981728.arc 0001_0000000072_1199981728.arc 0001_0000000074_1199981728.arc
```

/home/oracle/arch2:

```
0001_0000000069_1199981728.arc 0001_0000000071_1199981728.arc 0001_0000000073_1199981728.arc
0001_0000000070_1199981728.arc 0001_0000000072_1199981728.arc 0001_0000000074_1199981728.arc
```

```
SYS@orcl> select sequence#, name, FIRST_CHANGE#, NEXT_CHANGE# from v$archived_log;
```

SEQUENCE#	NAME	FIRST_CHANGE#	NEXT_CHANGE#
62	/home/oracle/arch1/0001_0000000062_1199981728.arc	2187169	2392382
62	/home/oracle/arch2/0001_0000000062_1199981728.arc	2187169	2392382
63	/home/oracle/arch1/0001_0000000063_1199981728.arc	2392382	2594334
63	/home/oracle/arch2/0001_0000000063_1199981728.arc	2392382	2594334
64	/home/oracle/arch1/0001_0000000064_1199981728.arc	2594334	2595377

64 /home/oracle/arch2/0001_0000000064_1199981728.arc	2594334	2595377
65 /home/oracle/arch1/0001_0000000065_1199981728.arc	2595377	2595380
65 /home/oracle/arch2/0001_0000000065_1199981728.arc	2595377	2595380
66 /home/oracle/arch1/0001_0000000066_1199981728.arc	2595380	2595387
66 /home/oracle/arch2/0001_0000000066_1199981728.arc	2595380	2595387
67 /home/oracle/arch1/0001_0000000067_1199981728.arc	2595387	2595390

SEQUENCE#	NAME	FIRST_CHANGE#	NEXT_CHANGE#
-----------	------	---------------	--------------

67 /home/oracle/arch2/0001_0000000067_1199981728.arc	2595387	2595390
68 /home/oracle/arch1/0001_0000000068_1199981728.arc	2595390	2698783
68 /home/oracle/arch2/0001_0000000068_1199981728.arc	2595390	2698783
69 /home/oracle/arch1/0001_0000000069_1199981728.arc	2698783	2700257
69 /home/oracle/arch2/0001_0000000069_1199981728.arc	2698783	2700257
70 /home/oracle/arch1/0001_0000000070_1199981728.arc	2700257	2700283
70 /home/oracle/arch2/0001_0000000070_1199981728.arc	2700257	2700283
71 /home/oracle/arch1/0001_0000000071_1199981728.arc	2700283	2700286
71 /home/oracle/arch2/0001_0000000071_1199981728.arc	2700283	2700286
72 /home/oracle/arch1/0001_0000000072_1199981728.arc	2700286	2700289
72 /home/oracle/arch2/0001_0000000072_1199981728.arc	2700286	2700289

SEQUENCE#	NAME	FIRST_CHANGE#	NEXT_CHANGE#
-----------	------	---------------	--------------

73 /home/oracle/arch1/0001_0000000073_1199981728.arc	2700289	2700292
73 /home/oracle/arch2/0001_0000000073_1199981728.arc	2700289	2700292
74 /home/oracle/arch1/0001_0000000074_1199981728.arc	2700292	2700295
74 /home/oracle/arch2/0001_0000000074_1199981728.arc	2700292	2700295

26 rows selected.

-- 물리적 삭제이나 논리적 삭제는 컨트롤파일 재생성, RMAN을 사용한 교차검증 방법으로 가능.

<<시나리오14>> 복구작업시 필요한 archive log file 삭제되었지만 redo log file이 있기때문에 문제없이 복구

```
SYS@orcl> select a.group#, b.sequence#, a.member, b.bytes/1024/1024 mb, b.status, b.archived, b.first_change#,
b.next_change# from v$logfile a, v$log b where a.group# = b.group# order by 1,2;
```

GROUP#	SEQUENCE#	MEMBER	MB	STATUS
ARCHIVED	FIRST_CHANGE#	NEXT_CHANGE#		
5	294	/u01/app/oracle/oradata/orcl/redo05.log	50	CURRENT
4133987	1.8447E+19			NO
6	292	/u01/app/oracle/oradata/orcl/redo06.log	50	INACTIVE
4133981	4133984			YES
7	293	/u01/app/oracle/oradata/orcl/redo07.log	50	INACTIVE
4133984	4133987			YES

장애 발생

```
SYS@orcl> select a.group#, b.sequence#, a.member, b.bytes/1024/1024 mb, b.status, b.archived, b.first_change#,
b.next_change# from v$logfile a, v$log b where a.group# = b.group# order by 1,2;
```

GROUP#	SEQUENCE#	MEMBER	MB
STATUS	ARC	FIRST_CHANGE#	NEXT_CHANGE#
1	73	/u01/app/oracle/oradata/orcl/redo01.log	20 INACTIVE
YES	2700289	2700292	
2	74	/u01/app/oracle/oradata/orcl/redo02.log	20 INACTIVE
YES	2700292	2700295	
3	75	/u01/app/oracle/oradata/orcl/redo03.log	20 CURRENT
NO	2700295	1.8447E+19	

```
SYS@orcl> !rm /home/oracle/arch2/0001_0000000074_1199981728.arc
```

```
SYS@orcl> !rm /u01/app/oracle/oradata/orcl/users01.dbf
```

```
SYS@orcl> !ls /home/oracle/arch*
```

```
/home/oracle/arch1:
```

```
0001_0000000069_1199981728.arc 0001_0000000071_1199981728.arc 0001_0000000073_1199981728.arc
0001_0000000070_1199981728.arc 0001_0000000072_1199981728.arc 0001_0000000074_1199981728.arc
```

/home/oracle/arch2:

0001_0000000069_1199981728.arc 0001_0000000071_1199981728.arc 0001_0000000073_1199981728.arc
0001_0000000070_1199981728.arc 0001_0000000072_1199981728.arc

복구진행

1. 테이블스페이스 offline

SYS@orcl> alter tablespace users offline immediate;
Tablespace altered.

2. 백업본 찾아 restore

SYS@orcl> ! cp -v /home/oracle/backup/open_bkp/users01.dbf /u01/app/oracle/oradata/orcl/
'/home/oracle/backup/open_bkp/users01.dbf' -> '/u01/app/oracle/oradata/orcl/users01.dbf'

3. 백업이후에 변경정보(리두) 적용

SYS@orcl> recover tablespace users
ORA-00279: change 2699840 generated at 05/12/2025 20:46:37 needed for thread 1
ORA-00289: suggestion : /home/oracle/arch2/0001_0000000069_1199981728.arc
ORA-00280: change 2699840 for thread 1 is in sequence #69

Specify log: {<RET>=suggested | filename | AUTO | CANCEL}
auto

ORA-00279: change 2700257 generated at 05/12/2025 20:52:22 needed for thread 1
ORA-00289: suggestion : /home/oracle/arch2/0001_0000000070_1199981728.arc
ORA-00280: change 2700257 for thread 1 is in sequence #70

ORA-00279: change 2700283 generated at 05/12/2025 20:53:23 needed for thread 1
ORA-00289: suggestion : /home/oracle/arch2/0001_0000000071_1199981728.arc
ORA-00280: change 2700283 for thread 1 is in sequence #71

ORA-00279: change 2700286 generated at 05/12/2025 20:53:27 needed for thread 1
ORA-00289: suggestion : /home/oracle/arch2/0001_0000000072_1199981728.arc
ORA-00280: change 2700286 for thread 1 is in sequence #72

Log applied.

Media recovery complete.

SYS@orcl>

4. 테이블스페이스 online

SYS@orcl> alter tablespace users online;

Tablespace altered.

SYS@orcl> select a.file#, b.name tbs_name, a.name file_name, a.checkpoint_change#, a.status from v\$datafile a, v\$tablespace b where a.ts# = b.ts#;

FILE#	TBS_NAME	FILE_NAME	CHECKPOINT_CHANGE#
1	SYSTEM	/u01/app/oracle/oradata/orcl/system01.dbf	2700295
2	UNDOTBS1	/u01/app/oracle/oradata/orcl/undotbs01	2700295
3	SYSAUX	/u01/app/oracle/oradata/orcl/sysaux01.dbf	2700295
7	USERS	/u01/app/oracle/oradata/orcl/users01.dbf	2704215

- 아카이브에 갭이 생기면 리두를 아카이브에 복사 가능
- 일반적으로 inactive상태의 redo log file을 복사하며 current redo log file은 복사하면 안됨.

[oracle@oel7v9r2 ~]\$ ls arch*

arch1:

```
0001_0000000062_1199981728.arc 0001_0000000065_1199981728.arc 0001_0000000068_1199981728.arc
0001_0000000063_1199981728.arc 0001_0000000066_1199981728.arc
0001_0000000064_1199981728.arc 0001_0000000067_1199981728.arc
```

arch2:

```
0001_0000000062_1199981728.arc 0001_0000000065_1199981728.arc 0001_0000000067_1199981728.arc
0001_0000000063_1199981728.arc 0001_0000000066_1199981728.arc 0001_0000000068_1199981728.arc
```

[oracle@oel7v9r2 ~]\$ ls arch*

arch1:

```
0001_0000000069_1199981728.arc 0001_0000000071_1199981728.arc 0001_0000000073_1199981728.arc
0001_0000000070_1199981728.arc 0001_0000000072_1199981728.arc 0001_0000000074_1199981728.arc
```

arch2:

```
0001_0000000069_1199981728.arc 0001_0000000071_1199981728.arc 0001_0000000073_1199981728.arc
0001_0000000070_1199981728.arc 0001_0000000072_1199981728.arc
[oracle@oel7v9r2 ~]$ cp -v arch1/0001_0000000074_1199981728.arc arch2/
'arch1/0001_0000000074_1199981728.arc' -> 'arch2/0001_0000000074_1199981728.arc'
```


<<시나리오15>> 복구작업시 필요한 archive log file이 손상되었을 경우 완전복구 실패로 인해 불완전 복구

```
SYS@orcl> alter system switch logfile;
```

System altered.

```
SYS@orcl> alter system switch logfile;
```

System altered.

```
SYS@orcl> alter system switch logfile;
```

System altered.

```
SYS@orcl> create table hr.emp tablespace users as select * from hr.employees;
```

Table created.

```
SYS@orcl> insert into hr.emp select * from hr.emp;
```

107 rows created.

```
SYS@orcl> /
```

214 rows created.

```
SYS@orcl> /
```

428 rows created.

```
SYS@orcl> /
```

856 rows created.

```
SYS@orcl> /
```

1712 rows created.

```
SYS@orcl> commit;
```

Commit complete.

```
SYS@orcl> select sequence#, name, FIRST_CHANGE#, NEXT_CHANGE# from v$archived_log;
```

SEQUENCE#	NAME	FIRST_CHANGE#	NEXT_CHANGE#

54	/home/oracle/arch1/arch_1_54_1200525254.arc	2886751	2888319
54	/home/oracle/arch2/arch_1_54_1200525254.arc	2886751	2888319
55	/home/oracle/arch1/arch_1_55_1200525254.arc	2888319	2890206
55	/home/oracle/arch2/arch_1_55_1200525254.arc	2888319	2890206
56	/home/oracle/arch1/arch_1_56_1200525254.arc	2890206	2897975
56	/home/oracle/arch2/arch_1_56_1200525254.arc	2890206	2897975
57	/home/oracle/arch1/arch_1_57_1200525254.arc	2897975	2998637
57	/home/oracle/arch2/arch_1_57_1200525254.arc	2897975	2998637
58	/home/oracle/arch1/arch_1_58_1200525254.arc	2998637	3001827
58	/home/oracle/arch2/arch_1_58_1200525254.arc	2998637	3001827
59	/home/oracle/arch1/arch_1_59_1200525254.arc	3001827	3001830
59	/home/oracle/arch2/arch_1_59_1200525254.arc	3001827	3001830
60	/home/oracle/arch1/arch_1_60_1200525254.arc	3001830	3001852
60	/home/oracle/arch2/arch_1_60_1200525254.arc	3001830	3001852

60 rows selected.

```
SYS@orcl> select a.file#, a.name, a.checkpoint_change#, b.status, b.change#, to_char(b.time, 'yyyy-mm-dd
hh24:mi:ss.sssss') time from v$datafile a, v$backup b where a.file# = b.file#;
```

FILE#	NAME	CHECKPOINT_CHANGE#	STATUS	TIME

1	/u01/app/oracle/oradata/orcl/system01.dbf	3001852	NOT ACTIVE	2897912 2025-05-12

23:27:25.

84445

3	/u01/app/oracle/oradata/orcl/sysaux01.dbf	3001852	NOT ACTIVE	2897912 2025-05-12
---	---	---------	------------	--------------------

23:27:25.

84445

```

4 /u01/app/oracle/oradata/orcl/undotbs01.dbf      3001852 NOT ACTIVE      2897912  2025-05-
12

```

```

23:27:25.

```

```

84445

```

```

FILE# NAME                                CHECKPOINT_CHANGE# STATUS
CHANGE# TIME
-----

```

```

7 /u01/app/oracle/oradata/orcl/users01.dbf      3001852 NOT ACTIVE      2897912  2025-05-
12

```

```

23:27:25.

```

```

84445

```

```

84445

```

```

SSYS@orcl> select a.group#, b.sequence#, a.member, b.bytes/1024/1024 mb, b.status, b.archived, b.first_change#,
b.next_change# from v$logfile a, v$log b where a.group# = b.group# order by 1,2;

```

```

GROUP# SEQUENCE# MEMBER                                MB STATUS
ARC FIRST_CHANGE# NEXT_CHANGE#
-----
1      61 /u01/app/oracle/oradata/orcl/redo01.log      10 CURRENT      NO
3001852 1.8447E+19
2      60 /u01/app/oracle/oradata/orcl/redo02.log      10 INACTIVE      YES
3001830 3001852
3      59 /u01/app/oracle/oradata/orcl/redo03.log      10 INACTIVE      YES
3001827 3001830

```

```

SYS@orcl> !ls -al /home/oracle/arch*

```

```

/home/oracle/arch1:

```

```

total 44824

```

```

drwxr-xr-x. 2 oracle dba      4096 May 13 00:28 .

```

```

drwx-----. 24 oracle dba    4096 May 10 23:58 ..

```

```

-rw-r-----. 1 oracle dba    1536 May 11 00:27 arch_1_50_1200525254.arc

```

```

-rw-r-----. 1 oracle dba    1024 May 11 00:27 arch_1_51_1200525254.arc

```

```
-rw-r-----. 1 oracle dba 10346496 May 12 23:22 arch_1_52_1200525254.arc
-rw-r-----. 1 oracle dba 10352128 May 12 23:22 arch_1_54_1200525254.arc
-rw-r-----. 1 oracle dba 10352128 May 12 23:22 arch_1_55_1200525254.arc
-rw-r-----. 1 oracle dba  9178624 May 12 23:29 arch_1_56_1200525254.arc
-rw-r-----. 1 oracle dba  210944 May 12 23:41 arch_1_57_1200525254.arc
-rw-r-----. 1 oracle dba  5424128 May 13 00:27 arch_1_58_1200525254.arc
-rw-r-----. 1 oracle dba    1024 May 13 00:27 arch_1_59_1200525254.arc
-rw-r-----. 1 oracle dba    3072 May 13 00:28 arch_1_60_1200525254.arc
```

/home/oracle/arch2:

total 44824

```
drwxr-xr-x. 2 oracle dba    4096 May 13 00:28 .
drwx-----. 24 oracle dba    4096 May 10 23:58 ..
-rw-r-----. 1 oracle dba    1536 May 11 00:27 arch_1_50_1200525254.arc
-rw-r-----. 1 oracle dba    1024 May 11 00:27 arch_1_51_1200525254.arc
-rw-r-----. 1 oracle dba 10346496 May 12 23:22 arch_1_52_1200525254.arc
-rw-r-----. 1 oracle dba 10352128 May 12 23:22 arch_1_54_1200525254.arc
-rw-r-----. 1 oracle dba 10352128 May 12 23:22 arch_1_55_1200525254.arc
-rw-r-----. 1 oracle dba  9178624 May 12 23:29 arch_1_56_1200525254.arc
-rw-r-----. 1 oracle dba  210944 May 12 23:41 arch_1_57_1200525254.arc
-rw-r-----. 1 oracle dba  5424128 May 13 00:27 arch_1_58_1200525254.arc
-rw-r-----. 1 oracle dba    1024 May 13 00:27 arch_1_59_1200525254.arc
-rw-r-----. 1 oracle dba    3072 May 13 00:28 arch_1_60_1200525254.arc
```

장애 발생

```
SYS@orcl> ! rm /u01/app/oracle/oradata/orcl/users01.dbf
```

```
SYS@orcl> !rm /home/oracle/arch1/arch_1_58_1200525254.arc /home/oracle/arch2/arch_1_58_1200525254.arc
```

```
SYS@orcl> !ls -al /home/oracle/arch*
```

복구

```
SYS@orcl> select status from v$instance;
```

STATUS

OPEN

리두 적용해야할 아카이브 파일이 없어서 완전 복구 실패
완전복구 실패시에는 불완전 복구 방식으로 수행해야 한다.

1. 데이터베이스 종료

```
SYS@orcl> shutdown abort
ORACLE instance shut down.
```

2. 모든 데이터파일을 restore

```
SYS@orcl> ! cp /home/oracle/backup/open_bkp/*.dbf /u01/app/oracle/oradata/orcl/
```

3. 데이터베이스 mount

```
SYS@orcl> startup mount
ORACLE instance started.
```

```
Total System Global Area 1543500144 bytes
Fixed Size                  8896880 bytes
Variable Size               905969664 bytes
Database Buffers           620756992 bytes
Redo Buffers                 7876608 bytes
Database mounted.
```

4. 데이터파일 상태 체크 - 해당 작업 체크가 안되면 복구 불가

```
SYS@orcl> select a.file#, b.name tbs_name, a.name file_name, a.checkpoint_change#, a.status from v$datafile a,
v$tablespace b where a.ts# = b.ts#;
```

FILE#	TBS_NAME	FILE_NAME
CHECKPOINT_CHANGE#	STATUS	

1	SYSTEM	/u01/app/oracle/oradata/orcl/system01.dbf
4134868	SYSTEM	
3	SYSAUX	/u01/app/oracle/oradata/orcl/sysaux01.dbf
4134868	ONLINE	
2	UNDOTBS1	/u01/app/oracle/oradata/orcl/undotbs01
4134868	ONLINE	

7 USERS
4135008 RECOVER

/u01/app/oracle/oradata/orcl/users01.dbf

-- 완전복구 수행하려고 시도했지만 불완전복구로 수행해야할 데이터파일을 꼭 online

SYS@orcl> alter database datafile 7 online;

Database altered.

5. 백업 컨트롤파일을 이용해서 cancel base recovery

SYS@orcl> recover database using backup controlfile until cancel;

ORA-00279: change 2897912 generated at 05/12/2025 23:27:25 needed for thread 1

ORA-00289: suggestion : /home/oracle/arch2/arch_1_56_1200525254.arc

ORA-00280: change 2897912 for thread 1 is in sequence #56

Specify log: {<RET>=suggested | filename | AUTO | CANCEL}

auto

ORA-00279: change 2897975 generated at 05/12/2025 23:29:18 needed for thread 1

ORA-00289: suggestion : /home/oracle/arch2/arch_1_57_1200525254.arc

ORA-00280: change 2897975 for thread 1 is in sequence #57

ORA-00278: log file '/home/oracle/arch2/arch_1_56_1200525254.arc' no longer needed for this recovery

ORA-00279: change 2998637 generated at 05/12/2025 23:41:45 needed for thread 1

ORA-00289: suggestion : /home/oracle/arch2/arch_1_58_1200525254.arc

ORA-00280: change 2998637 for thread 1 is in sequence #58

ORA-00278: log file '/home/oracle/arch2/arch_1_57_1200525254.arc' no longer needed for this recovery

ORA-00308: cannot open archived log '/home/oracle/arch2/arch_1_58_1200525254.arc'

ORA-27037: unable to obtain file status

Linux-x86_64 Error: 2: No such file or directory

Additional information: 7

SYS@orcl> recover database using backup controlfile until cancel;

ORA-00279: change 2998637 generated at 05/12/2025 23:41:45 needed for thread 1

ORA-00289: suggestion : /home/oracle/arch2/arch_1_58_1200525254.arc

ORA-00280: change 2998637 for thread 1 is in sequence #58

Specify log: {<RET>=suggested | filename | AUTO | CANCEL}

cancel

Media recovery cancelled.

SYS@orcl> alter database open resetlogs;

Database altered.

온라인(HOT) 백업

SYS@orcl> alter database begin backup;

Database altered.

SYS@orcl> ! cp -v /u01/app/oracle/oradata/orcl/*.dbf /home/oracle/backup/open_bkp/

'/u01/app/oracle/oradata/orcl/sysaux01.dbf' -> '/home/oracle/backup/open_bkp/sysaux01.dbf'

'/u01/app/oracle/oradata/orcl/system01.dbf' -> '/home/oracle/backup/open_bkp/system01.dbf'

'/u01/app/oracle/oradata/orcl/temp01.dbf' -> '/home/oracle/backup/open_bkp/temp01.dbf'

'/u01/app/oracle/oradata/orcl/undotbs01.dbf' -> '/home/oracle/backup/open_bkp/undotbs01.dbf'

'/u01/app/oracle/oradata/orcl/users01.dbf' -> '/home/oracle/backup/open_bkp/users01.dbf'

SYS@orcl> alter database end backup;

SYS@orcl> select a.file#, a.name, a.checkpoint_change#, b.status, b.change#, to_char(b.time, 'yyyy-mm-dd hh24:mi:ss.sssss') time from v\$datafile a, v\$backup b where a.file# = b.file#;

FILE#	NAME	CHECKPOINT_CHANGE#	STATUS	TIME
1	/u01/app/oracle/oradata/orcl/system01.dbf	2999573	NOT ACTIVE	2025-05-13 00:46:46.
02806				
3	/u01/app/oracle/oradata/orcl/sysaux01.dbf	2999573	NOT ACTIVE	2025-05-13 02:06:06.

00:46:46.

02806

4	/u01/app/oracle/oradata/orcl/undotbs01.dbf	2999573	NOT ACTIVE	2999573	2025-05-13
---	--	---------	------------	---------	------------

00:46:46.

02806

FILE#	NAME	CHECKPOINT_CHANGE#	STATUS	CHANGE#	TIME

7	/u01/app/oracle/oradata/orcl/users01.dbf	2999573	NOT ACTIVE	2999573	2025-05-13

00:46:46.

02806

SYS@orcl> select a.group#, b.sequence#, a.member, b.bytes/1024/1024 mb, b.status, b.archived, b.first_change#, b.next_change# from v\$logfile a, v\$log b where a.group# = b.group# order by 1,2;

GROUP#	SEQUENCE#	MEMBER	MB	STATUS
ARC FIRST_CHANGE#		NEXT_CHANGE#		

1	1	/u01/app/oracle/oradata/orcl/redo01.log	10	CURRENT
2998638	1.8447E+19			NO
2	0	/u01/app/oracle/oradata/orcl/redo02.log	10	UNUSED
0	0			YES
3	0	/u01/app/oracle/oradata/orcl/redo03.log	10	UNUSED
0	0			YES

SYS@orcl> alter system switch logfile;

System altered.

SYS@orcl> /

System altered.

SYS@orcl> /

System altered.

SYS@orcl> /

System altered.

SYS@orcl> select a.group#, b.sequence#, a.member, b.bytes/1024/1024 mb, b.status, b.archived, b.first_change#, b.next_change# from v\$logfile a, v\$log b where a.group# = b.group# order by 1,2;

GROUP#	SEQUENCE#	MEMBER	MB	STATUS	
	ARC	FIRST_CHANGE#	NEXT_CHANGE#		
1	4	/u01/app/oracle/oradata/orcl/redo01.log	10	INACTIVE	YES
2999714	2999717				
2	5	/u01/app/oracle/oradata/orcl/redo02.log	10	CURRENT	NO
2999717	1.8447E+19				
3	3	/u01/app/oracle/oradata/orcl/redo03.log	10	INACTIVE	YES
2999710	2999714				

SYS@orcl> alter database backup controlfile to '/home/oracle/backup/open_bkp/control_2025051301.ctl';

Database altered.

SYS@orcl> ! ls arch*

arch1:

```
arch_1_1_1200962609.arc  arch_1_4_1200962609.arc  arch_1_52_1200525254.arc  arch_1_56_1200525254.arc
arch_1_60_1200525254.arc
arch_1_2_1200962609.arc  arch_1_50_1200525254.arc  arch_1_54_1200525254.arc  arch_1_57_1200525254.arc
arch_1_61_1200525254.arc
arch_1_3_1200962609.arc  arch_1_51_1200525254.arc  arch_1_55_1200525254.arc  arch_1_59_1200525254.arc
```

arch2:

```
arch_1_1_1200962609.arc  arch_1_4_1200962609.arc  arch_1_52_1200525254.arc  arch_1_56_1200525254.arc
arch_1_60_1200525254.arc
arch_1_2_1200962609.arc  arch_1_50_1200525254.arc  arch_1_54_1200525254.arc  arch_1_57_1200525254.arc
```

arch_1_61_1200525254.arc

arch_1_3_1200962609.arc arch_1_51_1200525254.arc arch_1_55_1200525254.arc arch_1_59_1200525254.arc

<<시나리오 16>> 데이터파일 삭제, 모든 아카이브파일이 삭제된 경우 불완전한 복구(일관성 있는 백업, open, online, hot backup 있음)

1. 데이터베이스 종료

```
SYS@orcl> shutdown abort
ORACLE instance shut down.
```

2. 백업본 찾아 restore(control file, data file)

```
SYS@orcl> ! cp -v /home/oracle/backup/open_bkp/*.ctl,dbf /u01/app/oracle/oradata/orcl/
'backup/arch/open/control01_20250220.ctl' -> '/u01/app/oracle/oradata/orcl/control01_20250220.ctl'
'backup/arch/open/control01.ctl' -> '/u01/app/oracle/oradata/orcl/control01.ctl'
'backup/arch/open/control_20250224.ctl' -> '/u01/app/oracle/oradata/orcl/control_20250224.ctl'
'backup/arch/open/control_20250225.ctl' -> '/u01/app/oracle/oradata/orcl/control_20250225.ctl'
'backup/arch/open/sysaux01.dbf' -> '/u01/app/oracle/oradata/orcl/sysaux01.dbf'
'backup/arch/open/system01.dbf' -> '/u01/app/oracle/oradata/orcl/system01.dbf'
'backup/arch/open/temp01.dbf' -> '/u01/app/oracle/oradata/orcl/temp01.dbf'
'backup/arch/open/undotbs01.dbf' -> '/u01/app/oracle/oradata/orcl/undotbs01.dbf'
'backup/arch/open/users01.dbf' -> '/u01/app/oracle/oradata/orcl/users01.dbf'
```

3. 데이터베이스 mount

```
SYS@orcl> startup mount
ORACLE instance started.
```

```
Total System Global Area 1543500144 bytes
Fixed Size                  8896880 bytes
Variable Size               905969664 bytes
Database Buffers            620756992 bytes
Redo Buffers                 7876608 bytes
Database mounted.
```

4. 백업 컨트롤파일 이용해서 불완전 복구(cancel)

```
SYS@orcl> recover database using backup controlfile until cancel
ORA-00279: change 3381639 generated at 02/25/2025 14:00:02 needed for thread 1
ORA-00289: suggestion : /home/oracle/arch2/arch_1_274_1190473283.arc
ORA-00280: change 3381639 for thread 1 is in sequence #274
```

Specify log: {<RET>=suggested | filename | AUTO | CANCEL}

cancel

ORA-01547: warning: RECOVER succeeded but OPEN RESETLOGS would get error below

ORA-01195: online backup of file 1 needs more recovery to be consistent

ORA-01110: data file 1: '/u01/app/oracle/oradata/orcl/system01.dbf'

ORA-01112: media recovery not started

어떻게든 오픈하고자 resetlogs 시도

SYS@orcl> alter database open resetlogs;

alter database open resetlogs

*

ERROR at line 1:

ORA-01195: online backup of file 1 needs more recovery to be consistent

ORA-01110: data file 1: '/u01/app/oracle/oradata/orcl/system01.dbf'

백업 시점에 아카이브 없어서 미스매치

히든파라미터로 일관성 강제로 맞추기

SYS@orcl> select status from v\$instance;

STATUS

MOUNTED

SYS@orcl> select a.file#, b.name tbs_name, a.name file_name, a.checkpoint_change#, a.status from v\$datafile a,
v\$tablespace b where a.ts# = b.ts#;

FILE#	TBS_NAME	FILE_NAME
CHECKPOINT_CHANGE#	STATUS	

1	SYSTEM	/u01/app/oracle/oradata/orcl/system01.dbf
3381639	SYSTEM	
3	SYSAUX	/u01/app/oracle/oradata/orcl/sysaux01.dbf
3381639	ONLINE	
2	UNDOTBS1	/u01/app/oracle/oradata/orcl/undotbs01.dbf

3381639 ONLINE

7 USERS

/u01/app/oracle/oradata/orcl/users01.dbf

3381639 ONLINE

SYS@orcl> create pfile from spfile;

File created.

SYS@orcl> shutdown abort

ORACLE instance shut down.

SYS@orcl> ! vi \$ORACLE_HOME/dbs/initiorcl.ora

_allow_resetlogs_corruption=tur

-- 추가

SYS@orcl> startup pfile=\$ORACLE_HOME/dbs/initiorcl.ora mount

LRM-00105: 'tur' is not a legal Boolean for '_allow_resetlogs_corruption'

ORA-01078: failure in processing system parameters

<<시나리오17>> 데이터파일 삭제 모든 아카이브 파일이 삭제된 경우 불완전 복구(일관성있는 백업, close, offline, cold)

1. 데이터베이스 종료

```
SYS@orcl> shutdown abort
ORACLE instance shut down.
```

2. 백업본 찾아 restore(control file, data file)

```
SYS@orcl> ! cp backup/arch/open/*.ctl,dbf /u01/app/oracle/oradata/ORCL/
```

3. startup mount

```
SYS@orcl> startup mount
ORACLE instance started.
```

```
Total System Global Area 1543500144 bytes
Fixed Size                  8896880 bytes
Variable Size               905969664 bytes
Database Buffers           620756992 bytes
Redo Buffers                7876608 bytes
Database mounted.
```

4. 백업 컨트롤파일 이용해서 불완전 복구(cancel)

```
SYS@orcl> recover database using backup controlfile until cancel
ORA-00279: change 3278777 generated at 02/19/2025 14:16:38 needed for thread 1
ORA-00289: suggestion : /home/oracle/arch2/arch_1_272_1190473283.arc
ORA-00280: change 3278777 for thread 1 is in sequence #272
```

```
Specify log: {<RET>=suggested | filename | AUTO | CANCEL}
cancel
Media recovery cancelled.
```

5. 리셋로그로 오픈

```
SYS@orcl> alter database open resetlogs;
```

Database altered.

정보 확인

```
SYS@orcl> select a.group#, b.sequence#, a.member, b.bytes/1024/1024 mb, b.status, b.archived, b.first_change#,
b.next_change# from v$logfile a, v$log b where a.group# = b.group# order by 1,2;
```

GROUP#	SEQUENCE#	MEMBER	MB	STATUS
ARCHIVED	FIRST_CHANGE#	NEXT_CHANGE#		
5	1	/u01/app/oracle/oradata/ORCL/redo01.log	50	CURRENT
3278778	1.8447E+19			NO
6	0	/u01/app/oracle/oradata/ORCL/redo02.log	50	UNUSED
0	0			YES
7	0	/u01/app/oracle/oradata/ORCL/redo03.log	50	UNUSED
0	0			YES

```
SYS@orcl> select * from v$database_incarnation;
```

INCARNATION#	RESETLOGS_CHANGE#	RESETLOGS	PRIOR_RESETLOGS_CHANGE#	PRIOR_RES	STATUS
RESETLOGS_ID	PRIOR_INCARNATION#	FLASHBACK_DATABASE_ALLOWED	CON_ID		
1	1	17-APR-19	0		PARENT
1005785759	0	NO	0		
2	1920977	15-JAN-25	1	17-APR-19	PARENT
1190473283	1	NO	0		
3	3278778	25-FEB-25	1920977	15-JAN-25	CURRENT
1194018702	2	NO	0		

```
SYS@orcl> alter system archive log current;
```

System altered.

```
SYS@orcl> select sequence#, name, FIRST_CHANGE#, NEXT_CHANGE# from v$archived_log;
```

SEQUENCE#	NAME	FIRST_CHANGE#	NEXT_CHANGE#
270	/home/oracle/arch1/arch_1_270_1190473283.arc	3260541	3274563
270	/home/oracle/arch2/arch_1_270_1190473283.arc	3260541	3274563

271 /home/oracle/arch1/arch_1_271_1190473283.arc	3274563	3278265
271 /home/oracle/arch2/arch_1_271_1190473283.arc	3274563	3278265
1 /home/oracle/arch1/arch_1_1_1194018702.arc	3278778	3280083
1 /home/oracle/arch2/arch_1_1_1194018702.arc	3278778	3280083

6 rows selected.

-- 시퀀스 번호는 같을 수 있으나 뒤 resetlogs 번호가 달라진다.

-- resetlogs 번호는 select * from v\$database_incarnation;(위)의 RESETLOGS_ID 컬럼에서 확인

SYS@orcl> ! ls arch*

arch:

open

arch1:

arch_1_1_1194018702.arc arch_1_274_1190473283.arc arch_1_275_1190473283.arc arch_1_276_1190473283.arc

arch2:

arch_1_1_1194018702.arc arch_1_274_1190473283.arc arch_1_275_1190473283.arc arch_1_276_1190473283.arc

SYS@orcl> ! rm arch{1,2}/*1190473283.arc

SYS@orcl> ! ls arch*

arch:

open

arch1:

arch_1_1_1194018702.arc

arch2:

arch_1_1_1194018702.arc

-- 새 resetlogs로 인해 이전 아카이브파일은 필요없으니 삭제

online backup 받자

SYS@orcl> alter database begin backup;

Database altered.

SYS@orcl> ! cp /u01/app/oracle/oradata/ORCL/*.dbf /home/oracle/backup/open_bkp/


```
SYS@orcl> alter database end backup;
```

Database altered.

```
SYS@orcl> alter database backup controlfile to '/home/oracle/backup/open_bkp/control01.ctl';
```

Database altered.

```
SYS@orcl> alter system archive log current;
```

System altered.

정보 확인

```
SYS@orcl> select a.file#, a.name, a.checkpoint_change#, b.status, b.change#, to_char(b.time, 'yyyy-mm-dd
hh24:mi:ss.sssss') time from v$datafile a, v$backup b where a.file# = b.file#;
```

FILE#	NAME	CHECKPOINT_CHANGE#	STATUS
CHANGE#	TIME		

1	/u01/app/oracle/oradata/ORCL/system01.dbf	3280338 NOT ACTIVE	3280338
2025-02-25 16:03:43.57823			
2	/u01/app/oracle/oradata/ORCL/undotbs01.dbf	3280338 NOT ACTIVE	3280338
2025-02-25 16:03:43.57823			
3	/u01/app/oracle/oradata/ORCL/sysaux01.dbf	3280338 NOT ACTIVE	3280338
2025-02-25 16:03:43.57823			
7	/u01/app/oracle/oradata/ORCL/users01.dbf	3280338 NOT ACTIVE	3280338
2025-02-25 16:03:43.57823			

```
SYS@orcl> select a.group#, b.sequence#, a.member, b.bytes/1024/1024 mb, b.status, b.archived, b.first_change#,
b.next_change# from v$logfile a, v$log b where a.group# = b.group# order by 1,2;
```

GROUP#	SEQUENCE#	MEMBER	MB	STATUS
ARCHIVED	FIRST_CHANGE#	NEXT_CHANGE#		

5	1	/u01/app/oracle/oradata/ORCL/redo01.log	50 INACTIVE	YES
3278778	3280083			
6	2	/u01/app/oracle/oradata/ORCL/redo02.log	50 ACTIVE	YES
3280083	3280816			

7 3 /u01/app/oracle/oradata/ORCL/redo03.log 50 CURRENT NO
3280816 1.8447E+19

SYS@orcl> select sequence#, name, FIRST_CHANGE#, NEXT_CHANGE# from v\$archived_log;

SEQUENCE#	NAME	FIRST_CHANGE#	NEXT_CHANGE#
270	/home/oracle/arch1/arch_1_270_1190473283.arc	3260541	3274563
270	/home/oracle/arch2/arch_1_270_1190473283.arc	3260541	3274563
271	/home/oracle/arch1/arch_1_271_1190473283.arc	3274563	3278265
271	/home/oracle/arch2/arch_1_271_1190473283.arc	3274563	3278265
1	/home/oracle/arch1/arch_1_1_1194018702.arc	3278778	3280083
1	/home/oracle/arch2/arch_1_1_1194018702.arc	3278778	3280083
2	/home/oracle/arch1/arch_1_2_1194018702.arc	3280083	3280816 <-3280338
2	/home/oracle/arch2/arch_1_2_1194018702.arc	3280083	3280816

8 rows selected.

<<시나리오 18>> 데이터베이스가 정상적인 종료후에 inactive 상태에 있는 redo log file 삭제된 경우 복구

redo 체크

```
SYS@orcl> shutdown immediate
```

Database closed.

Database dismounted.

ORACLE instance shut down.

장애 발생

```
SYS@orcl> ! rm /u01/app/oracle/oradata/ORCL/redo01.log
```

```
SYS@orcl> startup
```

ORACLE instance started.

Total System Global Area 1543500144 bytes

Fixed Size 8896880 bytes

Variable Size 905969664 bytes

Database Buffers 620756992 bytes

Redo Buffers 7876608 bytes

Database mounted.

ORA-03113: end-of-file on communication channel

Process ID: 19079

Session ID: 15 Serial number: 20996

-- Alert log file 확인

```
SYS@orcl> ! ls /u01/app/oracle/oradata/ORCL/redo01.log
```

ls: cannot access /u01/app/oracle/oradata/ORCL/redo01.log: No such file or directory

```
SYS@orcl> conn / as sysdba
```

Connected to an idle instance.

복구

1. 데이터베이스 mount

```
SYS@orcl> startup mount
```

ORACLE instance started.

Total System Global Area 1543500144 bytes
 Fixed Size 8896880 bytes
 Variable Size 905969664 bytes
 Database Buffers 620756992 bytes
 Redo Buffers 7876608 bytes
 Database mounted.

2. redo 정보 확인한 후 문제가 발생한 리두 그룹을 삭제

```
SYS@orcl> select a.group#, b.sequence#, a.member, b.bytes/1024/1024 mb, b.status, b.archived, b.first_change#,
b.next_change# from v$logfile a, v$log b where a.group# = b.group# order by 1,2;
```

GROUP#	SEQUENCE#	MEMBER	MB	STATUS
ARCHIVED	FIRST_CHANGE#	NEXT_CHANGE#		
5	1	/u01/app/oracle/oradata/ORCL/redo01.log	50	INACTIVE
3278778	3280083			YES
6	2	/u01/app/oracle/oradata/ORCL/redo02.log	50	INACTIVE
3280083	3280816			YES
7	3	/u01/app/oracle/oradata/ORCL/redo03.log	50	CURRENT
3280816	1.8447E+19			NO

```
SYS@orcl> alter database drop logfile group 5;
```

Database altered.

```
SYS@orcl> select a.group#, b.sequence#, a.member, b.bytes/1024/1024 mb, b.status, b.archived, b.first_change#,
b.next_change# from v$logfile a, v$log b where a.group# = b.group# order by 1,2;
```

GROUP#	SEQUENCE#	MEMBER	MB	STATUS
ARCHIVED	FIRST_CHANGE#	NEXT_CHANGE#		
6	2	/u01/app/oracle/oradata/ORCL/redo02.log	50	INACTIVE
3280083	3280816			YES
7	3	/u01/app/oracle/oradata/ORCL/redo03.log	50	CURRENT
3280816	1.8447E+19			NO

3. 데이터베이스 open

```
SYS@orcl> alter database open;
```

Database altered.

4. 새로운 그룹 추가

```
SYS@orcl> alter database add logfile group 1 '/u01/app/oracle/oradata/ORCL/redo01.log' size 200m;
```

Database altered.

<<시나리오19>> 데이터베이스가 운영중에 inactive 상태에 있는 redo log file 삭제된 경우 복구

```
SYS@orcl> select a.group#, b.sequence#, a.member, b.bytes/1024/1024 mb, b.status, b.archived, b.first_change#,
b.next_change# from v$logfile a, v$log b where a.group# = b.group# order by 1,2;
```

GROUP#	SEQUENCE#	MEMBER	MB	STATUS	ARCHIVED	FIRST_CHANGE#	NEXT_CHANGE#
1	4	/u01/app/oracle/oradata/ORCL/redo01.log	200	CURRENT	NO		
3282691	1.8447E+19						
2	0	/u01/app/oracle/oradata/ORCL/redo02.log	50	UNUSED	YES		
0	0						
3	0	/u01/app/oracle/oradata/ORCL/redo03.log	50	UNUSED	YES		
0	0						
7	3	/u01/app/oracle/oradata/ORCL/redo07.log	50	INACTIVE	YES		
3280816	3282691						

장애 유발

```
SYS@orcl> ! rm /u01/app/oracle/oradata/ORCL/redo07.log
```

```
SYS@orcl> ! ls /u01/app/oracle/oradata/ORCL/redo07.log
```

```
ls: cannot access /u01/app/oracle/oradata/ORCL/redo07.log: No such file or directory
```

```
SYS@orcl> alter system switch logfile;
```

```
System altered.
```

```
SYS@orcl> /
```

```
System altered.
```

```
SYS@orcl> /
```

```
System altered.
```

```
SYS@orcl> /
```

```
System altered.
```

```
SYS@orcl> /
```

System altered.

```
SYS@orcl> /
```

System altered.

```
SYS@orcl> /
```

```
/ <- hang 발생
```

```
-- Alert log file확인
```

```
SYS@orcl> alter system switch logfile
```

```
# 정보 확인
```

```
SYS@orcl> select status from v$instance;
```

STATUS

OPEN

```
SYS@orcl> select a.group#, b.sequence#, a.member, b.bytes/1024/1024 mb, b.status, b.archived, b.first_change#,  
b.next_change# from v$logfile a, v$log b where a.group# = b.group# order by 1,2;
```

GROUP#	SEQUENCE#	MEMBER	MB	STATUS
ARCHIVED	FIRST_CHANGE#	NEXT_CHANGE#		
1	8	/u01/app/oracle/oradata/ORCL/redo01.log	200	INACTIVE
3282732	3282737			
2	9	/u01/app/oracle/oradata/ORCL/redo02.log	50	INACTIVE
3282737	3282741			
3	10	/u01/app/oracle/oradata/ORCL/redo03.log	50	CURRENT
3282741	1.8447E+19			
7	7	/u01/app/oracle/oradata/ORCL/redo07.log	50	INACTIVE
3282729	3282732			

```
SYS@orcl> ! ls arch*
```

redo log file을 삭제 후 재생성

```
SYS@orcl> alter database clear unarchived logfile group 7;
```

-- 아카이브 받지말고 지웠다가 다시 생성하기

Database altered.

```
SYS@orcl> ! ls /u01/app/oracle/oradata/ORCL/*.log
```

```
SYS@orcl> select a.group#, b.sequence#, a.member, b.bytes/1024/1024 mb, b.status, b.archived, b.first_change#,  
b.next_change# from v$logfile a, v$log b where a.group# = b.group# order by 1,2;
```

GROUP#	SEQUENCE#	MEMBER	MB	STATUS
ARCHIVED	FIRST_CHANGE#	NEXT_CHANGE#		

1	8	/u01/app/oracle/oradata/orcl/redo01.log	200	INACTIVE
3282732	3282737			YES
2	9	/u01/app/oracle/oradata/orcl/redo02.log	50	INACTIVE
3282737	3282741			YES
3	10	/u01/app/oracle/oradata/orcl/redo03.log	50	CURRENT
3282741	1.8447E+19			NO
7	0	/u01/app/oracle/oradata/orcl/redo07.log	50	UNUSED
3282729	328273			YES